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Anomaly detection using user behavioral biometrics profiling method and apparatus

Abstract

Techniques for determining anomalous user behavior in connection with an online application are disclosed. In one embodiment, a method is disclosed comprising obtaining user behavior data in connection with a user of an application, generating feature data using the obtained user behavior data, obtaining one or more user behavior anomaly predictions from one or more anomaly prediction models trained to output a user behavior anomaly prediction in response to the feature data. Each user behavior anomaly prediction indicates a probability that the user behavior is anomalous. A user behavior anomaly determination is made using the user behavior anomaly prediction(s).

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Background/Summary

BACKGROUND INFORMATION

(1) Fraudulent user activity represents significant issues to entities, including entities conducting business online, online retailers, ecommerce platform providers providing goods and services to online users, etc.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 provides an example illustrating anomalous user behavior prediction in accordance with one or more embodiments of the present disclosure;
- (2) FIG. 2 provides examples of user behavior data in accordance with one or more embodiments;
- (3) FIG. 3 provides an illustrative model prediction generation example using multiple prediction models in accordance with one or more embodiments of the present disclosure.;
- (4) FIG. 4 provides a user behavior data examples and corresponding user behavior anomaly determination examples in accordance with one or more embodiments of the present disclosure;
- (5) FIG. 5 provides an exemplary illustration of a user behavior anomaly determination publication in accordance with one or more embodiments of the present disclosure;
- (6) FIG. 6 provides an example of a user behavior anomaly process flow in accordance with one or more embodiments of the present disclosure;
- (7) FIG. 7 provides an example of a model training process flow used in accordance with one or more embodiments of the present disclosure;
- (8) FIG. 8 is a schematic diagram illustrating an example of a network within which the systems and methods disclosed herein could be implemented according to some embodiments of the present disclosure;
- (9) FIG. 9 depicts is a schematic diagram illustrating an example of client device in accordance with some embodiments of the present disclosure; and
- (10) FIG. 10 is a block diagram illustrating the architecture of an exemplary hardware device in accordance with one or more embodiments of the present disclosure.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

- (11) The disclosed embodiments can be used in detecting anomalies in user behavior indicative of a bad actor with a fraudulent or other malevolent purpose. Some examples of fraudulent behavior include credit card fraud (e.g., using stolen credit card information to purchase goods and/or services), account takeover fraud (e.g., stealing an existing customer's login credentials to access the customer's online account), malicious bots (e.g., using automated tools to access existing customer accounts and/or create new accounts) and promotion fraud (e.g., reusing promotional codes, abusing coupon policies, etc.).
- (12) A bad actor operating online can be anonymous and can operate from virtually any location and at any time. As such, the bad actor can operate without being detected for some period of time, during which the bad actor can cause significant financial loss, computer security breaches, digital data breaches, user account breaches, malware installation, computing system and computer networking attacks, etc.
- (13) Embodiments of the present disclosure provide a mechanism for detecting and reporting anomalous behavior (e.g., of a bad actor) in an expedient manner. Embodiments of the present disclosure provide an effective mechanism for avoiding damage by the bad actors to their targets, such as and without limitation online providers (e.g., ecommerce providers), online users, computing systems and computing resources, electronic networks, etc.
- (14) An actor's behavior provides valuable insight into the intent or purpose of the actor, e.g., whether or not the actor is a bad actor. In accordance with embodiments of the present disclosure, a user's behavior can be evaluated to determine anomalous behavior, e.g., behavior that is

inconsistent with a user's previous behavior, behavior that is inconsistent with other user's behavior, etc. By way of a non-limiting example, the user behavior can comprise behavior, or activity, associated with an online provider of goods and/or services.

(15) In accordance with at least one disclosed embodiment, a user's behavior in connection with one or more channels (e.g., online, offline or some combination of online and offline channels), including an application, such as without limitation a web application, a mobile application, a chatbot or the like, can be evaluated and used in determining whether or not the user behavior is anomalous, inconsistent, user behavior. In accordance with one or more embodiments, one or more statistical models can be used to make an anomalous behavior prediction, which can be used to make an anomalous behavior determination. By way of a non-limiting example, an anomalous behavior determination can be used by a system (e.g., a risk mitigation system, such as and without limitation a fraud detection and prevention system of an online provider) to take action to mitigate risk.

(16) A mechanism is provided for detecting anomalous user behavior using a user behavior anomaly prediction generated by an anomaly prediction model trained to output the prediction in response to feature data generated from user behavior data obtained in connection with one or more applications. Each user behavior anomaly prediction indicates a probability that a user's behavior is anomalous (e.g., different from the user's past behavior, different from other users' behavior, etc.). As a user is interacting with an application, user behavior data is collected and used to identify anomalous user behavior. Since anomalous user behavior can be detected as a user is interacting with an application, action can be taken in a timely manner to minimize damage resulting from the anomalous user behavior.

(17) FIG. 1 provides an example illustrating anomalous user behavior prediction and determination in accordance with one or more embodiments of the present disclosure. User behavior data can be received from a number of channels **102**. By way of some non-limiting examples, a channel **102** can be an application executing on a client device (e.g., mobile, desktop, laptop, etc. device), an application accessible via an electronic network (e.g., an application executing on a remote device, such as a server or other computing device), or some combination thereof. By way of a further non-limiting example, the channel(s) **102** can be used by users of an online provider to access (e.g., purchase) goods and/or services of the online provider. Some non-limiting examples of user behavior include keyboard input, cursor activity, touch screen usage, mobile device sensor input, etc.

(18) In accordance with one or more embodiments, user behavior data from the channel(s) **102** can be received by user behavior data receiver **104**, which can be configured to receive and store user behavior data in the user behavior data store(s) **106**. By way of a non-limiting examples, a user behavior data receiver **104** can be a component of a web application, an application executing on a client device or some combination thereof.

(19) In accordance with one or more embodiments, user behavior data can be associated with a session. As used herein, a session can be a group of interactions with a user (e.g., via a channel **102**) in a given time frame, where the time frame can be a defined amount of time (e.g., 30 minutes). In accordance with one or more embodiments, user behavior data received by the user behavior data receiver(s) **104** can be stored in the user behavior data store **106** and associated with (in the store **106**) an identifier (e.g., session identifier) identifying the session from which the user behavior data is received. In accordance with one or more embodiments, user behavior data from the user behavior data store **106** can be input to data aggregator and feature generator **108**.

(20) FIG. 2 provides examples of user behavior data that can be stored in behavior data store **106** and can be input to data aggregator and feature generator **108** in accordance with one or more embodiments. In the example shown in FIG. 2, web application **202** and client application **204** can each be an example of a channel **102**. Blocks **206** and **208** include examples of user behavior data that can be received (e.g., by user behavior data receiver(s) **104**) from web application **202** and

client application **204**, respectively.

(21) In the example shown in FIG. 2, blocks **206** and **208** include user behavior data items A, B, D and E. Keyboard usage under item A includes user behavior data, such as and without limitation, usage of left and/or right shift keys, keyboard shortcuts, functions keys, left and right control keys, keypad, etc. Keyboard usage can also include data about the extent to which a user uses cut and paste operations during a session. By way of a non-limiting example, cut and paste operations can signal that a user is copying information (e.g., someone else's information) into a form rather than providing the information from the user's own memory. Keyboard usage data can be compared with keyboard usage data from one or more previous sessions to determine whether keyboard usage is the same or different from the previous session(s).

(22) Internet Protocol (IP) related features under item B includes IP-related data, such as and without limitation a network address of a client device being used by a user. A current session's IP-related data can be compared with IP-related data from one or more previous sessions to determine whether the current session's IP-related data matches the IP-related data from the previous session(s).

(23) Existing Customer (Past Web Usage Behavioral Pattern—Funnel under item D includes previous behavioral pattern data (e.g., user behavior data associated with one or more previous sessions). Bot/Automated JavaScript (JS) usage under item E includes information identifying whether any scripts are used by the user. By way of a non-limiting example, user behavior data under item E can indicate use, by a user, of a script to automatically fill in input fields (e.g., using information from an application such as a word processing or spreadsheet application).

(24) In the example shown in FIG. 2, block **206** also includes item C and block **208** includes items C and F. Cursor activity under item C of block **206** includes cursor movements (e.g., mouse, pointer, etc. input). This category of user behavior data can include data indicating mouse button usage and left and/or right hand input. Touch screen usage under item C of block **208** includes touch screen usage data, such as and without limitation use of an onscreen keyboard, left hand and/or right hand use, screen coordinates corresponding to onscreen button selections, etc.

(25) Device sensor information under item F of block **208** can include device (e.g., mobile device) sensor information indicating how the user is holding the device, such as and without limitation at a certain horizontal and/or vertical angle, flat on a surface, landscape or portrait mode, etc. Item F user behavior information can further include information indicating whether the user is on the phone during the session. By way of a non-limiting example, the user might be on the phone with a customer service representative who is providing the user with assistance in filling out an online form.

(26) In accordance with one or more embodiments, user behavior data from blocks **206** and **208** can be used by data aggregator and feature generator **108** to generate additional (e.g., aggregate) user behavior information (e.g., for a given session, or sessions), such as the examples provided at block **210** of FIG. 2.

(27) In the example of FIG. 2, user behavior information under items 1-4, 13 and 15 of block **210** can be generated using user behavior information under item A of block **206** and/or block **208**. Funnel navigation pattern under item 1 can include user behavior information such as navigation pattern information (e.g., a navigation path used by the user to arrive at a certain point, such as an application user interface screen or web page). By way of a further non-limiting example, a funnel navigation pattern can refer to a navigation path taken by a user to reach an intended outcome or action (e.g., making a purchase, signing up for a newsletter, playing content, etc.)

(28) Pause length under item 2 can include user behavior information indicating a length of a pause by the user, e.g., an aggregate amount of time that the user took between actions. Session length under item 3 can include user behavior information indicating a length of a session, e.g., as an aggregate amount of time of a session. Shortcut usage under item 4 can include information indicating the shortcut(s) used by the user during the session. Preference of interaction under item

13 can include user behavior information indicating interaction preferences of the user. By way of a non-limiting example, user behavior information under item 13 can include information indicating which channel(s) **102** the user prefers using in connection with one or more activities (e.g., a user's preferred channel **102** for checking bills). Forms edit count under item 15 can include user behavior information indicating a number of forms edited by the user during a session, or sessions.

(29) In the example of FIG. 2, script usage under item 5 of block **210** can be generated using user behavior information under item E of block **206** and/or block **208** and can include information indicating whether or not the user uses a script (e.g., JavaScript), or other programming, to fill in information (e.g., from another application such as a spreadsheet, word processing, or the like). Fill form behavior under item 6 of block **210** can be generated using user behavior information under item D of block **206** and/or block **208** and can include user behavior information indicating whether or not the user entered information (e.g., the user's name, address, etc.) from memory or with a pasting operation (e.g., copy and paste, cut and paste, etc.) when providing requested information.

(30) Pointed mouse movement under item 7 of block **210** of the example can be generated using user behavior information under item E of block **206** and/or block **208** and can include user behavior information indicating whether or not mouse movements are automated (e.g., mouse movement using an emulator such as Selenium to emulate a user's interaction with a browser, application user interface, etc.). API calls under item 8 of block **210** can be generated using user behavior information under items C and D of block **206** and/or item D of block **208** and can include information indicating a number of application programming interface (API) calls. A higher than normal number of API calls can be an indicator of anomalous behavior.

(31) Local storage data check under item 9 of block **210** can be generated using user behavior information under item E of block **206** and/or block **208** and can include user behavior information indicating whether or not a browser cookie or local data storage has been altered or enabled (or put) on the user's computing device without authorization and/or can indicate that browser cookie or local data storage contents cannot be authenticated. Aimless mouse movements under item 10 of block **210** can be generated using user behavior information under item C of block **206** and/or block **208** and can include information indicating whether or not mouse input of the user is aimless (e.g., not directed, not purposeful, etc.)

(32) Device gyroscope parameters under item 11 of block **210** can be generated using device (e.g., mobile device, phone, etc.) sensor information under item F of block **208** and can include information indicating a rotation of the user's computing device. By way of non-limiting example, rotational information can be used with other item F user behavior information such as accelerometer information indicating an orientation of the user's computing device. Button press length under item 12 of block **210** can be generated using touch screen usage information under item C of block **208** and can include user behavior information indicating a length of time that the user maintained pressure on a button.

(33) Preference of interaction information under item 13 of block **210** can be generated using user behavior information under items A and C of block **206** and/or block **208** and can include user behavior information indicating a user's interaction preferences. By way of a non-limiting example, a user's interaction preferences can indicate what application(s) the user prefers to use, such as which application the user typically uses to check the user's bills. Information comprehension time under item 14 of block **210** can be generated using user behavior information under item C of block **206** and/or block **208** and can include user behavior information indicating how much time a user uses to review displayed information, such as information displayed in a web page or an application user interface display.

(34) Forms edit count under item 15 of block **210** can be generated using user behavior information under item A of block **206** and/or block **208** and can include user behavior information indicating the number of edits made by the user to a form, or forms. Assistance used under item 17 of block

210 can be generated using device sensor information under item F of block **208** and can include user behavior information indicating whether the user used assistance (e.g., telephone support) in filling a form or navigating a user interface. IP region, state, IsProxy, IsVPN, IsCloudInfra, IP organization under item 18 of block **210** can be generated using user behavior information under item B of block **206** and/or block **208** and/or item C of block **206** and can include information indicating an IP region, a state, information indicating whether or not any proxy is being used, information indicating whether or not the IP address being used is behind a virtual private network (VPN), server identity information (e.g., organization, region, state), etc.

(35) Blocks **206**, **208** and **210** are inexhaustive examples of user behavior information that can be used with one or more disclosed embodiments. Other information indicative of a user's behavior can also be used. In addition, and while many of the items in block **210** can be generated using information of the type associated with items in blocks **206** and **208**, age/gender comparison information under items in block **210** can be generated from other sources. For example, user behavior information under item 16 of block **210** can come from user account information. User behavior information under item 16 can include information indicating a user's age and gender and comparisons with other ages and genders.

(36) Referring again to FIG. 1, user behavior information, or data, such as that discussed in connection with FIG. 2 can be used to generate a user behavior feature data superset at data aggregator and feature generator block **108**. At data preprocessing **110**, the user behavior feature data superset can undergo various preprocessing steps, such as and without limitation, a variance check, correlation analysis and data normalization, prior to being input to feature selection **112**. A variance check can provide a measure of the variability of the user behavior feature data. Correlation analysis can be used to identify correlations among the user behavior feature data. Normalization can be used to normalize the user behavior feature data. By way of a non-limiting example, normalization can be used to normalize the user behavior feature data to have the same standard deviation.

(37) At feature selection **112**, at least a portion of the user behavior feature data superset generated at block **108** and preprocessed at block **110** is selected to be used as the user behavior feature data that is used as prediction model input **114**.

(38) Block **110** can use principal component analysis (PCA) to select the user behavior feature data (e.g., from the user behavior feature data superset) to be used as prediction model input **114**. Feature selection **112** using PCA can be used to select a set of features from the user behavior feature data superset that best represents the behavior of the user being analyzed. By way of some non-limiting examples, PCA can be used to select a set of features having a certain level of variability in accordance with the variance analysis and a certain level of correlations in the user behavior data determined using correlations analysis. The feature data in the set of selected features can be normalized using data normalization performed at data preprocessing block **110**.

(39) In the example shown in FIG. 1, in accordance with at least one embodiment, prediction model input **114** comprising the user behavior data selected at feature selection **112** becomes input to user behavior anomaly prediction model(s) **116**. The user behavior anomaly prediction model(s) **116** uses the prediction model input **114** to generate one or more user behavior anomaly prediction(s) **120** in connection with a user and a session. By way of one non-limiting example, the user behavior anomaly prediction(s) **120** can comprise a number of predictions, each of which can comprise a probability, or likelihood, that the user behavior is anomalous.

(40) At user behavior anomaly determination **122**, the user behavior anomaly prediction(s) **120** can be used to make a determination whether or not the user's behavior is anomalous (different, inconsistent, irregular, etc.) as compared to the user's past behavior and/or the current and/or past behavior of other users. In accordance with one or more embodiments, the user behavior anomaly determination **122** can be published by user behavior anomaly determination publisher **124**. By way of one non-limiting example, the publisher **124** can alert a system (e.g., a risk mitigation

system of an online provider) by communicating information identifying a user and session and an indicator that the user's behavior is anomalous or not anomalous. By way of another non-limiting example, the publisher **124** can comprise a data store (e.g., database or other data store) and an application programming interface (API), which can be used by a system to retrieve a user behavior anomaly determination **122** for a user, or users.

(41) By way of a non-limiting example, the channel(s) **102** can be provided by an e-commerce provider providing goods and/or services to users, the user behavior anomaly determination **122** can be used by a computing system (e.g., a risk mitigation system, such as fraud detection and prevention system) to identify potential undesirable activity by users or entities impersonating users—e.g., a bad actor.

(42) In accordance with one or more embodiments, user behavior anomaly prediction model(s) **116** can comprise a number of different prediction models. FIG. **3** provides an illustrative model prediction generation example using multiple prediction models in accordance with one or more embodiments of the present disclosure.

(43) In accordance with one or more embodiments, user behavior anomaly prediction model(s) **116** can comprise more than one model, which is shown as models **302-1**, **302-2**, . . . , **302-N** in the example of FIG. **3**. For ease with referencing, models **302-1**, **302-2**, . . . , **302-N** are also collectively referred to herein as models **302**. As illustrated in the example, prediction model input **114** can be input to each of the models **302**. Each of the models **302** can generate a user behavior anomaly prediction **120**, which becomes input to model output aggregator **304**. Model output aggregator **304** receives the user behavior anomaly prediction **120** from each of the models **302** and generates an aggregate model prediction **306**. Aggregate model prediction **306** can be used to make a user behavior anomaly determination **122**.

(44) In accordance with one or more embodiments, models **302** can comprise an Isolation Forest machine learning algorithm, a variational auto-encoder and a one-class support vector machine (SVM). An isolation forest algorithm is an unsupervised machine learning algorithm that can be used to isolate anomalies (or outliers) in the prediction model input **114** feature data by recursively generating partitions on the prediction model input **114** comprising the user behavior feature set. Isolation forest can isolate outliers by randomly selecting a feature from the prediction model input **114** and then randomly selecting a split value between maximum and minimum values of the selected feature. The random partitioning of features can be used to distinguish between anomalous user behavior feature data values and normal user behavior feature data values. The isolation forest machine learned model can provide a probability score indicating a likelihood that the user behavior represented by the prediction model input **114** is anomalous user behavior.

(45) Another example of a model that can be included in the models **302** is a variational auto-encoder (VAE), which is a deep learning technique. A VAE is a generative model and is a type of autoencoder which generates encoded data by encoding input data (e.g., prediction model input **114**), decodes the encoded data and determines differences between the input data and the decoded data. Unlike other autoencoders, a VAE's encoder encodes the input data into a latent space, and its decoder decodes data points randomly sampled from the latent space to generate the decoded data that is used to determine whether or not the VAE was able to replicate the input data. An inability of the VAE to replicate the input data (e.g., the input data is not replicated by the encoding and decoding done by the VAE) indicates an anomaly. The VAE can provide a probability of the likelihood that the user behavior represented by the prediction model input **114** is anomalous user behavior.

(46) A one-class SVM is another example of a model that can be included in the models **302**. One-class SVM is an unsupervised learning algorithm. A one-class SVM model can be trained to identify anomalous user behavior by clustering the prediction model input **114** and then concentrating on any of the prediction model input **114** data identified as outliers to determine whether or not prediction model input **114** represents anomalous user behavior. The one-class SVM

can provide a probability score indicating a likelihood that the user behavior represented by the prediction model input **114** includes at least one anomalous user behavior.

(47) By way of a non-limiting example, the probability output from a model **302** can be a value in the range of 0-1, with 0 indicating the least likelihood of anomalous user behavior and 1 indicating the highest likelihood of anomalous user behavior.

(48) The probability scores received from each of the models **302** can be aggregated by model output aggregator **304**. By way of a non-limiting example, the model output aggregator **304** can average the output received from the models **302** to generate the aggregate model prediction **306**. Any type of aggregation of the output of the models **302** can be used in connection with disclosed embodiments.

(49) In aggregating the probabilities received from the models **302**, the model output aggregator **304** can treat the output of each of the models **302** equally or use a weighting that results in the model output with a greater weighting having more influence in determining the aggregate model prediction **306**. For example, a 25% (or 0.25) weighting might be associated with two of the models **302** and a 50% (or 0.5) weighting might be associated with a third model **302**. In this case, the probability output by the third model **302** would have a greater influence on the aggregate model prediction **306** than the probability output by either of the other two models **302**. In the example, the third model has as much influence on the aggregate model prediction **306** as the other two models combined.

(50) To generate a weighted probability for a model **302**, model output aggregator **304** can multiply the probability output by the model **302** by the weight associated with the model **302**. The model output aggregator **304** can then aggregate (e.g., average) the weighted probabilities of each model **302** to generate the aggregate model prediction **306**.

(51) In accordance with one or more embodiments, user behavior data store **106** (of FIG. 1) can store real time user behavior data that can be used to determine whether anomalous user behavior exists in connection with an active session. User behavior anomaly determination publisher **124** can communicate a determination made in connection with an active session to risk mitigation system (e.g., a fraud detection and prevention system) so that the system can address the anomalous user behavior.

(52) FIG. 4 provides a user behavior data examples and corresponding user behavior anomaly determination examples in accordance with one or more embodiments of the present disclosure. In the example shown in FIG. 4, table **400** includes examples of user behavior information, or data, that can be received by user behavior data receiver(s) **104** from channel(s) **102**. Table **400** can be stored in user behavior data store **106**. A session and user combination can have considerably more user behavior data than that shown in table **400**. The user behavior data shown in table **400** (e.g., page ID, time taken, click count, key press queue) is provided for illustrative purposes only.

(53) Table **420** can be stored in a data store, as is discussed in more detail in connection with FIG. 5. Table **420** provides an example of user behavior anomaly determinations **122** that can be made using the user behavior data in Table **400**. In the example shown in FIG. 4, although each user and session combination has one associated user behavior anomaly determination **122**, each user and session combination can have multiple user behavior anomaly determinations **122**, each of which can have a corresponding timestamp indicating the point in the session at which the determination is made. By way of a non-limiting example, the multiple determinations **122** can be used to establish a user behavior anomaly determination pattern, which can be used (e.g., by a risk mitigation system) to determine whether or not to take some type of action, e.g., one or more actions to mitigate the risk.

(54) Referring to Table **400**, each of rows **402**, **404** and **406** is associated with a session and user. Each session and each user is identified by a unique identifier. In the example of FIG. 4, a current user interface display is identified by a page identifier, or page ID. Although not shown in the example of FIG. 4, table **400** can include an identifier of each page visited by a user in a current

session, or sessions. By way of a non-limiting example, the page ID of each page visited by a user in a session can be used in determining a navigation pattern (e.g., user behavior information under item 1 of block **210** of FIG. 2).

(55) In the example shown in FIG. 4, each row **402**, **404** and **406** includes a time taken column, which can include information indicating the amount of time taken on the current page identified by the page ID (e.g., user behavior information under item 2 of block **210** of FIG. 2). Although not shown in the example, a time taken can be associated with each page ID of each page visited by a user in a current session, and an aggregate of the times that the user has taken at each page in a session can be used to determine a session time (e.g., user behavior information under item 3 of block **210** of FIG. 2).

(56) Each of rows **402**, **404** and **406** of table **400** includes a count of the number of clicks (e.g., mouse clicks) by the user in the current session. A click count is a type of cursor activity (e.g., user behavior information under item C of block **206** or block **208** of FIG. 2). By way of a non-limiting example, click count can be used in determining user behavior information under items 9, 10, 12, 13 and/or 14 of block **210** of FIG. 2.

(57) In the example shown in FIG. 4, table **400** includes a key press queue, which can be processed to generate the user behavior information under items in blocks **206**, **208** and **210** of FIG. 2.

(58) Table **420** includes a user behavior anomaly determination **122** for each user and session combination shown in table **400**. In accordance with embodiments of the present disclosure, a user behavior anomaly determination **122** can be triggered in various ways, such as and without limitation triggered based on an interval of time (e.g., an increment of time in a session, since the last determination, etc.), triggered based on the amount of user behavior data received, etc. In accordance with one or more embodiments, at the start of a session, a default can be used that indicates that the user behavior is not anomalous prior to an initial user behavior anomaly determination **122** being triggered. In accordance with one or more embodiments, an initial user behavior anomaly determination **122** may be made sooner relative to subsequent user behavior anomaly determinations **122** made in the session.

(59) In the example shown in row **406** of table **400**, assuming that the current page is the first page in the session, the session has just begun and there is minimal information collected in connection with the session at this point. In such a case, the associated user behavior anomaly determination **122** in row **426** corresponding to row **406** (of table **400**) can have a default, initial value of “No.”

(60) As shown in the example of FIG. 4, each session ID and user ID combination can have an associated user behavior anomaly determination **122** indicating whether or not anomalous user behavior has been detected in connection with the user and session combination. In the example shown in table **420**, row **424** indicates that anomalous user behavior has been detected in connection with the session and user. By way of a non-limiting example, using the user behavior example of table **400**, the determination might be made based on the lengthy time taken, and the user's information pasting (i.e., Ctrl V) behavior.

(61) In the example shown in FIG. 4, the session and user associated with row **422** indicates that there is no anomalous user behavior detected in the user behavior (e.g., user behavior indicated in row **402** of table **400**).

(62) In accordance with one or more embodiments, a user behavior anomaly determination **122** corresponding to a user and session can be stored for access by a risk mitigation system. FIG. 5 provides an exemplary illustration of a user behavior anomaly determination publication in accordance with one or more embodiments of the present disclosure.

(63) In the example shown in FIG. 5, user behavior anomaly determination publisher **124** comprises an anomaly determination data store **502**, an application programming interface (API) **504** and user behavior anomaly determination cache **506**. In accordance with one or more embodiments, anomaly determination data store **502** can be a database management system's database comprising a number of tables. The database can include one or more tables associating

user information (e.g., name, address, age, gender, other demographic information, etc.), session information (date and time commenced, data and time ended, network information, such as network address, user interface navigation path, user input, etc.) and one or more user behavior anomaly determinations **122** determined for the user and session combination. In accordance with one or more embodiments, each user behavior anomaly determination **122** can be associated with a timestamp (e.g., a data and time) indicating a timing of the determination, a user (e.g., a user identifier, or userID) and session (e.g., a session identifier, or sessionID).

(64) As shown in the example, a risk mitigation system **508** can communicate with the user behavior anomaly determination publisher **124** and access user behavior anomaly determinations **122** stored in anomaly determination data store **502**. In accordance with one or more embodiments, API **504** can be used by risk mitigation system **508** to access user behavior anomaly determinations **122** stored in anomaly determination data store **502**.

(65) In addition to anomaly determination data store **502**, the example of FIG. 5 includes a user behavior anomaly determination cache **506**, which can store some of the user behavior anomaly determinations **122** stored in the anomaly determination data store **502**. By way of a non-limiting example, user behavior anomaly determination cache **506** might store one or more weeks of the most recent user behavior anomaly determinations **122** for a number of user and session combinations.

(66) By way of a non-limiting example, the user behavior anomaly determination publisher **124** can receive a user behavior anomaly determination access request (e.g., a query) from the risk mitigation system **508** via the API **504**. The user behavior anomaly determination request might identify one or more user and session combinations and a time period. In response to a user behavior anomaly determination request, the user behavior anomaly determination publisher **124** can retrieve the requested user behavior anomaly determination(s) **122** and store them in the user behavior anomaly determination cache **506** for access by the risk mitigation system **508** (e.g., via the API **504**).

(67) In accordance with one or more embodiments, the user behavior anomaly determination publisher **124** can be hosted by a cloud computing platform provider (e.g., Amazon Web Services, etc.) Alternatively, the user behavior anomaly determination publisher **124** can be hosted by a server computing device. As yet another alternative, the user behavior anomaly determination publisher **124** can be hosted by some combination of a cloud computing platform provider and a server computing device. As yet another alternative, the user behavior anomaly determination publisher **124** can be hosted by one or more computing devices configured to execute one or more of blocks **104**, **108**, **110**, **112**, **116** and **122**.

(68) FIG. 6 provides an example of a user behavior anomaly process flow in accordance with one or more embodiments of the present disclosure. Briefly, steps included in process flow **600** can be performed in connection with user behavior and interaction with an online or offline channel (e.g., an online or offline application, chatbot, mobile application, etc.). It should be apparent that any channel (online or offline) can be used in connection with the embodiments disclosed herein. In accordance with one or more embodiments, process flow **600** can operate to make a user behavior anomaly determination using one or more user behavior anomaly predictions and communicate the determination to another system, such as a risk mitigation system, which can determine what if any action to take in connection with the user behavior anomaly determination.

(69) At step **602**, user behavior information, or data, is obtained. By way of a non-limiting example, the user behavior data that is collected can correspond to a particular user and session. By way of a further non-limiting example, the session can be a session with an online ecommerce provider (e.g., a provider of goods, services or some combination) associated with a risk mitigation system. By way of a non-limiting example, the user behavior data can be obtained via a channel **102** and user behavior data receiver **104**. By way of a further non-limiting example, the user behavior data can be retrieved from a user behavior data store **106**.

(70) In accordance with one or more embodiments, the user behavior data that is obtained at step **602** can be in connection with any user behavior involving a user's interaction with a channel **102** being used or accessed by the user.

(71) As discussed herein (e.g., in connection with FIGS. **2** and **4**), the user behavior data obtained in connection with a user and session can comprise a number of different types of data, as discussed in connection with FIG. **2**, such as and without limitation keyboard usage, touch screen usage, historical user behavior data, network address and device sensor data. In addition, and as discussed herein, the user behavior can include aggregate data, such as that discussed in connection with block **210** of FIG. **2**.

(72) At step **604**, model input can be generated. As discussed in connection with FIG. **1**, data aggregator and feature generator **108** can use user behavior data from user behavior data store **106** to generate user behavior feature data using user behavior data and/or aggregate user behavior data (e.g., such as the data shown in blocks **206**, **208** and **210** of FIG. **2**). Step **604** can include one or more data preprocessing operations, such as and without limitation a variance check, correlation analysis and data normalization. As discussed herein in connection with FIG. **1**, data preprocessing can be performed by data processing module **110** of FIG. **1**.

(73) In accordance with one or more embodiments, the model input generation (at step **604**) can include feature selections to select at least a portion of the user behavior feature data to be used as the model input. By way of a non-limiting example, PCA can be used to select from a user behavior feature data superset a set of features that best represents the behavior of the user whose behavior is being analyzed to identify any anomalous behavior. In accordance with one or more embodiments, feature selection can be used to select a set of features having a certain level of variability in accordance with the variance analysis and a certain level of correlations in the user behavior data determined using correlations analysis. The feature data in the set of selected features can be normalized using data normalization. As discussed herein, variance analysis, correlation analysis and data normalization can be performed at data preprocessing **110**.

(74) At step **606**, one or more user behavior anomaly prediction(s) **120** are obtained. In accordance with one or more embodiments, the model input generated at step **604** can be input to one or more user behavior anomaly prediction models, such as models **302**. As discussed in connection with FIG. **3**, models **302** can comprise an Isolation Forest model, a VAE model and a one-class SVM model.

(75) As discussed herein, an isolation forest model can be trained to isolate outlying feature data from the model input by randomly selecting a feature from the model input and then randomly selecting a split value between maximum and minimum values of the selected feature. The random partitioning of features can be used to distinguish between anomalous data values and normal data values. A VAE model can be trained to identify anomalous user behavior by attempting to replicate the model input using an encoder to encode some or all of the model input data, using a decoder to decode the encoded data and then comparing the original model input data with the decoded data to identify anomalies in the user behavior model input. An SVM model can be trained to identify anomalous user behavior by clustering the model input and then concentrating on any of the model input data identified as outliers to determine whether or not prediction model input represents anomalous user behavior.

(76) In accordance with one or more embodiments, each of the models **302** used at step **606** can provide a probability score indicating a likelihood that the user behavior represented by the prediction model input **114** (or user behavior feature data input to model(s) **116**) is anomalous user behavior.

(77) At step **608**, a user behavior anomaly determination **122** is made. In accordance with one or more embodiments, step **608** can include a step of aggregating multiple user behavior anomaly predictions to yield one probability score in the case that multiple user behavior anomaly predictions are obtained at step **606**. In any case, the user behavior anomaly prediction(s) obtained

at step **606** is/are used to make the user behavior anomaly determination **122** at step **608**.

(78) In accordance with one or more embodiments, the user behavior anomaly determination **122** can be made using a probability score (e.g., a probability score output by a single model or an aggregate probability score(s) output by more than one model) and a threshold probability value. The threshold probability can be empirically determined based on the number of true and false positives.

(79) By way of a non-limiting example, assuming that a higher probability score is more indicative of anomalous user behavior than a lower probability score, the probability threshold can be set such that a probability score that is equal to or greater than threshold probability is indicative of anomalous user behavior. As yet another example, a lower probability score can be more indicative of anomalous user behavior, the probability threshold can be set such that a probability score that is equal to or less than threshold probability is indicative of anomalous user behavior. Generally, the user behavior anomaly determination can be made based on whether or not the probability satisfies the threshold.

(80) At step **610**, the user behavior anomaly determination can be communicated to a computing system, such as a risk mitigation system. By way of a non-limiting example, the user behavior anomaly determination can be communicated to an ecommerce platform provider, an entity conducting business online, etc. that is providing channels **102** offering goods and/or services to online customers. The user behavior anomaly determination(s) communicated at step **610** can be used to identify a bad actor such that the risk can be mitigated.

(81) In accordance with one or more embodiments, a user behavior anomaly determination communicated (at step **610**) to a system (e.g., a risk mitigation system **508**) can result in the system monitoring a user session, terminating the session, flagging the session, raising an alert, etc.

(82) As discussed herein in connection with one or more embodiments, one or more user behavior anomaly prediction models (e.g., models **302**) can be trained to predict anomalous user behavior. FIG. 7 provides an example of a model training process flow used in accordance with one or more embodiments of the present disclosure. In general, steps included in process flow **700** can be performed using user behavior data obtained in connection with a number of sessions and users accessing one or more channels (online and/or offline channels) provided by one or more providers. In accordance with one or more embodiments, process flow **700** can operate to use the user behavior data to generate a number of training examples that can be used to train each user behavior anomaly prediction model using one or more machine learning algorithms.

(83) At step **702**, a corpus of user behavior data can be obtained (e.g., from user behavior data store **106**). In accordance with one or more embodiments, the corpus of user behavior data can be associated with multiple users and sessions. By way of a non-limiting example, user behavior data obtained at step **702** in connection with multiple sessions and multiple users of one or more channels can be stored (or logged) and used as the corpus of activity data.

(84) At step **704**, training data is generated. In accordance with one or more embodiments, the training data comprising a number of feature datasets can be generated using some or all of the user behavior data obtained at step **702**.

(85) At step **706**, one or more models are trained using the training data. By way of a non-limiting example, training data comprising a number of feature datasets generated (at step **704**) using some or all of the corpus of user behavior data (obtained at step **702**) can be used to train the models **302**. In accordance with one or more embodiments, the training data can comprise multiple training examples (e.g., multiple feature datasets) that can be used to train each of the user behavior anomaly prediction models **302** to make a user behavior anomaly prediction **120** in response to feature dataset input generated using user behavior data associated with a user and a current session with a channel **102**. As discussed, each model (e.g., each of the models **302**) can be trained to output a probability score indicating a likelihood that the user behavior represented by the prediction model input **114** is anomalous user behavior. In accordance with one or more

embodiments, **706** can be used with one or more machine learning algorithms and training data generated (at step **704**) using data obtained (at step **702**) to train models **302**.

(86) In accordance with one or more embodiments, user behavior can be monitored to detect anomalous user behavior using one or more trained user behavior anomaly predictions models **302** and user behavior data, such as that discussed in connection with FIGS. **2** and **4**. Unlike a brick-and-mortar establishment, an online provider (e.g., an ecommerce provide, online retailer, or any entity doing business and/or providing goods and/or services online) is unable to assess someone (e.g., a user, customer, etc.) in person, obtain someone's credentials for verification purposes, etc. Online fraud is increasing dramatically. Advantageously, embodiments of the present disclosure can be used to make user behavior anomaly determinations, and then communicate the user behavior anomaly determinations to a risk mitigation system which can take steps to mitigate the risk.

(87) FIG. **8** provides an example of components of a general environment in accordance with one or more embodiments. FIG. **8** shows components of a general environment in which the systems and methods discussed herein may be practiced. Not all the components may be required to practice the disclosure, and variations in the arrangement and type of the components may be made without departing from the spirit or scope of the disclosure. As shown, system **800** of FIG. **8** includes local area networks (“LANs”)/wide area networks (“WANs”)—network **805**, wireless network **810**, mobile devices (client devices) **802-804** and client device **801**. FIG. **8** additionally includes a server **808**. Examples of web servers include without limitation, application servers, content servers, search servers, advertising servers, etc.

(88) In accordance with one or more embodiments, server **808** can include functionality disclosed herein in connection with one or more embodiments. Server **808** can host one or more web applications, for which user reaction is being monitored.

(89) One embodiment of mobile devices **802-804** is described in more detail below. Generally, however, mobile devices **802-804** may include virtually any portable computing device capable of receiving and sending a message over a network, such as network **805**, wireless network **810**, or the like. Mobile devices **802-804** may also be described generally as client devices that are configured to be portable. Thus, mobile devices **802-804** may include virtually any portable computing device capable of connecting to another computing device and receiving information. Such devices include multi-touch and portable devices such as, cellular telephones, smart phones, display pagers, radio frequency (RF) devices, infrared (IR) devices, Personal Digital Assistants (PDAs), handheld computers, laptop computers, wearable computers, smart watch, tablet computers, phablets, integrated devices combining one or more of the preceding devices, and the like. As such, mobile devices **802-804** typically range widely in terms of capabilities and features. For example, a cell phone may have a numeric keypad and a few lines of monochrome LCD display on which only text may be displayed. In another example, a web-enabled mobile device may have a touch sensitive screen, a stylus, and an HD display in which both text and graphics may be displayed.

(90) A web-enabled mobile device may include a browser application that is configured to receive and to send web pages, web-based messages, and the like. The browser application may be configured to receive and display graphics, text, multimedia, and the like, employing virtually any web based language, including a wireless application protocol messages (WAP), and the like. In one embodiment, the browser application is enabled to employ Handheld Device Markup Language (HDML), Wireless Markup Language (WML), WMLScript, JavaScript, Standard Generalized Markup Language (SMGL), HyperText Markup Language (HTML), eXtensible Markup Language (XML), and the like, to display and send a message.

(91) Mobile devices **802-804** also may include at least one client application that is configured to receive content from another computing device. The client application may include a capability to provide and receive textual content, graphical content, audio content, and the like. The client application may further provide information that identifies itself, including a type, capability, name,

and the like. In one embodiment, mobile devices **802-804** may uniquely identify themselves through any of a variety of mechanisms, including a phone number, Mobile Identification Number (MIN), an electronic serial number (ESN), or other mobile device identifier.

(92) In some embodiments, mobile devices **802-804** may also communicate with non-mobile client devices, such as client device **801**, or the like. Client device **801** may include virtually any computing device capable of communicating over a network to send and receive information. The set of such devices may include devices that typically connect using a wired or wireless communications medium such as personal computers, multiprocessor systems, microprocessor-based or programmable consumer electronics, network PCs, or the like. Thus, client device **801** may also have differing capabilities for displaying navigable views of information.

(93) Client device **801** and mobile devices **801-804** may be capable of sending or receiving signals, such as via a wired or wireless network, or may be capable of processing or storing signals, such as in memory as physical memory states, and may, therefore, operate as a server. Thus, devices capable of operating as a server may include, as examples, dedicated rack-mounted servers, desktop computers, laptop computers, set top boxes, integrated devices combining various features, such as two or more features of the foregoing devices, or the like.

(94) Wireless network **810** is configured to couple mobile devices **802-804** and its components with network **805**. Wireless network **810** may include any of a variety of wireless sub-networks that may further overlay stand-alone ad-hoc networks, and the like, to provide an infrastructure-oriented connection for mobile devices **802-804**. Such sub-networks may include mesh networks, Wireless LAN (WLAN) networks, cellular networks, and the like.

(95) Network **805** is configured to communicatively couple web server **808** with other computing devices, including, client device **801**, and through wireless network **810** to mobile devices **802-804**. Network **805** is enabled to employ any form of computer readable media for communicating information from one electronic device to another. Also, network **805** can include the Internet in addition to local area networks (LANs), wide area networks (WANs), direct connections, such as through a universal serial bus (USB) port, other forms of computer-readable media, or any combination thereof. On an interconnected set of LANs, including those based on differing architectures and protocols, a router acts as a link between LANs, enabling messages to be sent from one to another, and/or other computing devices.

(96) Within the communications networks utilized or understood to be applicable to the present disclosure, such networks will employ various protocols that are used for communication over the network. Signal packets communicated via a network, such as a network of participating digital communication networks, may be compatible with or compliant with one or more protocols. Signaling formats or protocols employed may include, for example, TCP/IP, UDP, QUIC (Quick UDP Internet Connection), DECnet, NetBEUI, IPX, APPLE TALK™, or the like. Versions of the Internet Protocol (IP) may include IPv4 or IPv6. The Internet refers to a decentralized global network of networks. The Internet includes local area networks (LANs), wide area networks (WANs), wireless networks, or long haul public networks that, for example, allow signal packets to be communicated between LANs. Signal packets may be communicated between nodes of a network, such as, for example, to one or more sites employing a local network address. A signal packet may, for example, be communicated over the Internet from a user site via an access node coupled to the Internet. Likewise, a signal packet may be forwarded via network nodes to a target site coupled to the network via a network access node, for example. A signal packet communicated via the Internet may, for example, be routed via a path of gateways, servers, etc. that may route the signal packet in accordance with a target address and availability of a network path to the target address.

(97) A server, such as server **808**, may be capable of sending or receiving signals, such as via a wired or wireless network, or may be capable of processing or storing signals, such as in memory as physical memory states. Devices capable of operating as a server may include, as examples,

dedicated rack-mounted servers, desktop computers, laptop computers, set top boxes, integrated devices combining various features, such as two or more features of the foregoing devices, or the like. Servers may vary widely in configuration or capabilities, but generally, a server may include one or more central processing units and memory. A server may also include one or more mass storage devices, one or more power supplies, one or more wired or wireless network interfaces, one or more input/output interfaces, or one or more operating systems, such as Windows Server, Mac OS X, Unix, Linux, FreeBSD, or the like.

(98) In some embodiments, users are able to access services provided by servers, such as web server **808** as well as other servers, such as and without limitation authentication servers, search servers, email servers, social networking services servers, SMS servers, IM servers, MMS servers, exchange servers, photo-sharing services servers, and travel services servers, via the network **805** using their various devices **801-804**. In some embodiments, application server can host applications, such as an e-commerce application, a search engine, a content recommendation and/or distribution application, etc.

(99) In some embodiments, web server **808** can store various types of applications and application related information including application data. As is discussed in more detail below, examples of application data include user behavior, application behavior, page visitation sequences, and visit intent and action data. In accordance with some embodiments, web server **808** can host an application, or applications, embodying functionality described herein.

(100) Moreover, although FIG. **8** illustrates web server **808** as single computing devices, respectively, the disclosure is not so limited. For example, one or more functions of web server **808** may be distributed across one or more distinct computing devices. Moreover, in one embodiment, web server **808** may be integrated into a single computing device, without departing from the scope of the present disclosure.

(101) FIG. **9** is a schematic diagram illustrating an example embodiment of a computing device that may be used within the present disclosure. Device **900** may include many more or less components than those shown in FIG. **9**. However, the components shown are sufficient to disclose an illustrative embodiment for implementing the present disclosure. Device **900** may represent, for example, client device **801** and mobile devices **801-804** discussed above in relation to FIG. **8**.

(102) As shown in the figure, device **900** includes a processing unit (CPU) **922** in communication with a mass memory **930** via a bus **924**. Device **900** also includes a power supply **926**, one or more network interfaces **950**, an audio interface **952**, a display **954**, a keypad **956**, an illuminator **958**, an input/output interface **960**, a haptic interface **962**, an optional global positioning systems (GPS) receiver **964** and a camera(s) or other optical, thermal or electromagnetic sensors **966**. Device **900** can include one camera/sensor **966**, or a plurality of cameras/sensors **966**, as understood by those of skill in the art. The positioning of the camera(s)/sensor(s) **966** on device **900** can change per device **900** model, per device **900** capabilities, and the like, or some combination thereof.

(103) Device **900** may optionally communicate with a base station (not shown), or directly with another computing device. Network interface **950** includes circuitry for coupling device **900** to one or more networks, and is constructed for use with one or more communication protocols and technologies as discussed above.

(104) Optional GPS transceiver **964** can determine the physical coordinates of device **900** on the surface of the Earth, which typically outputs a location as latitude and longitude values. GPS transceiver **964** can also employ other geo-positioning mechanisms, including, but not limited to, triangulation, assisted GPS (AGPS), E-OTD, CI, SAI, ETA, BSS or the like, to further determine the physical location of device **900** on the surface of the Earth. In an embodiment device **900** may through other components, provide other information that may be employed to determine a physical location of the device, including for example, a MAC address, Internet Protocol (IP) address, or the like.

(105) Mass memory **930** includes a RAM **932**, a ROM **934**, and other storage means. Mass

memory **930** illustrates another example of computer storage media for storage of information such as computer readable instructions, data structures, program modules or other data. Mass memory **930** stores a basic input/output system (“BIOS”) **940** for controlling low-level operation of device **900**. The mass memory also stores an operating system **941** for controlling the operation of device **900**. It will be appreciated that this component may include a general purpose operating system such as a version of UNIX, or LINUX™, or a specialized client communication operating system such as Windows Client™, or the Symbian® operating system. The operating system may include, or interface with a Java virtual machine module that enables control of hardware components and/or operating system operations via Java application programs.

(106) Memory **930** further includes one or more data stores, which can be utilized by device **900** to store, among other things, applications **942** and/or other data. For example, data stores may be employed to store information that describes various capabilities of device **900**. The information may then be provided to another device based on any of a variety of events, including being sent as part of a header during a communication, sent upon request, or the like. At least a portion of the capability information may also be stored on a disk drive or other storage medium (not shown) within device **900**.

(107) Applications **942** may include computer executable instructions which, when executed by device **900**, transmit, receive, and/or otherwise process audio, video, images, and enable telecommunication with a server and/or another user of another client device. Other examples of application programs or “apps” in some embodiments include browsers, calendars, contact managers, task managers, transcoders, photo management, database programs, word processing programs, security applications, spreadsheet programs, games, search programs, and so forth. Applications **942** may further include search client **945** that is configured to send, to receive, and/or to otherwise process a search query and/or search result using any known or to be known communication protocols. Although a single search client **945** is illustrated it should be clear that multiple search clients may be employed. For example, one search client may be configured to enter a search query message, where another search client manages search results, and yet another search client is configured to manage serving advertisements, IMs, emails, and other types of known messages, or the like.

(108) As shown in FIG. **10**, internal architecture **1000** of a computing device(s), computing system, computing platform, user devices, set-top box, smart TV and the like includes one or more processing units, processors, or processing cores, (also referred to herein as CPUs) **1012**, which interface with at least one computer bus **1002**. Also interfacing with computer bus **1002** are computer-readable medium, or media, **1006**, media disk interface **1008**, network interface **1014**, memory **1004**, e.g., random access memory (RAM), run-time transient memory, read only memory (ROM), media disk drive interface **1020** as an interface for a drive that can read and/or write to media, display interface **1010** as interface for a monitor or other display device, keyboard interface **1016** as interface for a keyboard, pointing device interface **1018** as an interface for a mouse or other pointing device, and miscellaneous other interfaces **1022** not shown individually, such as parallel and serial port interfaces and a universal serial bus (USB) interface.

(109) Memory **1004** interfaces with computer bus **1002** so as to provide information stored in memory **1004** to CPU **1012** during execution of software programs such as an operating system, application programs, device drivers, and software modules that comprise program code, and/or computer executable process steps, incorporating functionality described herein, e.g., one or more of process flows described herein. CPU **1012** first loads computer executable process steps from storage, e.g., memory **1004**, computer readable storage medium/media **1006**, removable media drive, and/or other storage device. CPU **1012** can then execute the stored process steps in order to execute the loaded computer-executable process steps. Stored data, e.g., data stored by a storage device, can be accessed by CPU **1012** during the execution of computer-executable process steps.

(110) Persistent storage, e.g., medium/media **1006**, can be used to store an operating system and

one or more application programs. Persistent storage can further include program modules and data files used to implement one or more embodiments of the present disclosure, e.g., listing selection module(s), targeting information collection module(s), and listing notification module(s), the functionality and use of which in the implementation of the present disclosure are discussed in detail herein.

(111) Network link **1034** typically provides information communication using transmission media through one or more networks to other devices that use or process the information. For example, network link **1034** may provide a connection through local network **1024** to a host computer **1026** or to equipment operated by a Network or Internet Service Provider (ISP) **1030**. ISP equipment in turn provides data communication services through the public, worldwide packet-switching communication network of networks now commonly referred to as the Internet **1032**.

(112) A computer called a server host **1034** connected to the Internet **1032** hosts a process that provides a service in response to information received over the Internet **1032**. For example, server host **1034** hosts a process that provides information representing video data for presentation at a display via display interface **1010**. It is contemplated that the components of system **1000** can be deployed in various configurations within other computer systems, e.g., host and server.

(113) At least some embodiments of the present disclosure are related to the use of computer system **1000** for implementing some or all of the techniques described herein. According to one embodiment, those techniques are performed by computer system **1000** in response to processing unit **1012** executing one or more sequences of one or more processor instructions contained in memory **1004**. Such instructions, also called computer instructions, software and program code, may be read into memory **1004** from another computer-readable medium **1006** such as storage device or network link. Execution of the sequences of instructions contained in memory **1004** causes processing unit **1012** to perform one or more of the method steps described herein. In alternative embodiments, hardware, such as ASIC, may be used in place of or in combination with software. Thus, embodiments of the present disclosure are not limited to any specific combination of hardware and software, unless otherwise explicitly stated herein.

(114) The signals transmitted over network link and other networks through communications interface, carry information to and from computer system **1000**. Computer system **1000** can send and receive information, including program code, through the networks, among others, through network link and communications interface. In an example using the Internet, a server host transmits program code for a particular application, requested by a message sent from computer, through Internet, ISP equipment, local network and communications interface. The received code may be executed by processor **1012** as it is received, or may be stored in memory **1004** or in a storage device or other non-volatile storage for later execution, or both.

(115) The present disclosure has been described with reference to the accompanying drawings, which form a part hereof, and which show, by way of non-limiting illustration, certain example embodiments. Subject matter may, however, be embodied in a variety of different forms and, therefore, covered or claimed subject matter is intended to be construed as not being limited to any example embodiments set forth herein; example embodiments are provided merely to be illustrative. Likewise, a reasonably broad scope for claimed or covered subject matter is intended. Among other things, for example, the subject matter may be embodied as methods, devices, components, or systems. Accordingly, embodiments may, for example, take the form of hardware, software, firmware, or any combination thereof (other than software per se). The following detailed description is, therefore, not intended to be taken in a limiting sense.

(116) Throughout the specification and claims, terms may have nuanced meanings suggested or implied in context beyond an explicitly stated meaning. Likewise, the phrase “in some embodiments” as used herein does not necessarily refer to the same embodiment, and the phrase “in another embodiment” as used herein does not necessarily refer to a different embodiment. It is intended, for example, that claimed subject matter include combinations of example embodiments

in whole or in part.

(117) In general, terminology may be understood at least in part from usage in context. For example, terms such as “and,” “or,” or “and/or,” as used herein may include a variety of meanings that may depend at least in part upon the context in which such terms are used. Typically, “or” if used to associate a list, such as A, B, or C, is intended to mean A, B, and C, here used in the inclusive sense, as well as A, B or C, here used in the exclusive sense. In addition, the term “one or more” as used herein, depending at least in part upon context, may be used to describe any feature, structure, or characteristic in a singular sense or may be used to describe combinations of features, structures, or characteristics in a plural sense. Similarly, terms, such as “a,” “an,” or “the,” again, may be understood to convey a singular usage or to convey a plural usage, depending at least in part upon context. In addition, the term “based on” may be understood as not necessarily intended to convey an exclusive set of factors and may, instead, allow for the existence of additional factors not necessarily expressly described, again, depending at least in part on context.

(118) The present disclosure has been described with reference to block diagrams and operational illustrations of methods and devices. It is understood that each block of the block diagrams or operational illustrations, and combinations of blocks in the block diagrams or operational illustrations, can be implemented by means of analog or digital hardware and computer program instructions. These computer program instructions can be provided to a processor of a general purpose computer to alter its function as detailed herein, a special purpose computer, ASIC, or other programmable data processing apparatus, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, implement the functions/acts specified in the block diagrams or operational block or blocks. In some alternate implementations, the functions/acts noted in the blocks can occur out of the order noted in the operational illustrations. For example, two blocks shown in succession can in fact be executed substantially concurrently or the blocks can sometimes be executed in the reverse order, depending upon the functionality/acts involved.

(119) For the purposes of this disclosure, a non-transitory computer-readable medium (or computer-readable storage medium/media) stores computer data, which data can include computer program code (or computer-executable instructions) that is executable by a computer, in machine-readable form. By way of example, and not limitation, a computer-readable medium may comprise computer-readable storage media, for tangible or fixed storage of data, or communication media for transient interpretation of code-containing signals. Computer-readable storage media, as used herein, refers to physical or tangible storage (as opposed to signals) and includes without limitation volatile and non-volatile, removable and non-removable media implemented in any method or technology for the tangible storage of information such as computer-readable instructions, data structures, program modules or other data. Computer-readable storage media can tangibly encode computer-executable instructions that when executed by a processor associated with a computing device perform functionality disclosed herein in connection with one or more embodiments.

(120) Computer-readable storage media includes, but is not limited to, RAM, ROM, EPROM, EEPROM, flash memory or other solid-state memory technology, CD-ROM, DVD, or other optical storage, cloud storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other physical or material medium which can be used to tangibly store thereon the desired information or data or instructions and which can be accessed by a computer or processor.

(121) For the purposes of this disclosure a module is a software, hardware, or firmware (or combinations thereof) system, process or functionality, or component thereof, that performs or facilitates the processes, features, and/or functions described herein (with or without human interaction or augmentation). A module can include sub-modules. Software components of a module may be stored on a computer readable medium for execution by a processor. Modules may be integral to one or more servers, or be loaded and executed by one or more servers. One or more

modules may be grouped into an engine or an application.

(122) For the purposes of this disclosure the term “user”, “subscriber” “consumer” or “customer” should be understood to refer to a user of an application or applications as described herein and/or a consumer of data supplied by a data provider. By way of example, and not limitation, the term “user” or “subscriber” can refer to a person who receives data provided by the data or service provider over the Internet in a browser session, or can refer to an automated software application which receives the data and stores or processes the data.

(123) Those skilled in the art will recognize that the methods and systems of the present disclosure may be implemented in many manners and as such are not to be limited by the foregoing exemplary embodiments and examples. In other words, functional elements being performed by single or multiple components, in various combinations of hardware and software or firmware, and individual functions, may be distributed among software applications at either the client level or server level or both. In this regard, any number of the features of the different embodiments described herein may be combined into single or multiple embodiments, and alternate embodiments having fewer than, or more than, all of the features described herein are possible.

(124) Functionality may also be, in whole or in part, distributed among multiple components, in manners now known or to become known. Thus, myriad software/hardware/firmware combinations are possible in achieving the functions, features, interfaces and preferences described herein. Moreover, the scope of the present disclosure covers conventionally known manners for carrying out the described features and functions and interfaces, as well as those variations and modifications that may be made to the hardware or software or firmware components described herein as would be understood by those skilled in the art now and hereafter.

(125) Furthermore, the embodiments of methods presented and described as flowcharts in this disclosure are provided by way of example in order to provide a more complete understanding of the technology. The disclosed methods are not limited to the operations and logical flow presented herein. Alternative embodiments are contemplated in which the order of the various operations is altered and in which sub-operations described as being part of a larger operation are performed independently.

(126) In the preceding specification, various example embodiments have been described with reference to the accompanying drawings. However, it will be evident that various modifications and changes may be made thereto, and additional embodiments may be implemented without departing from the broader scope of the disclosed embodiments as set forth in the claims that follow. The specification and drawings are accordingly to be regarded in an illustrative rather than restrictive sense.

Claims

1. A method comprising: obtaining, at a computing device, user behavior data in connection with a user of an application; detecting, by the computing device, an action by another user related to an interaction with the application; generating, by the computing device, user behavior feature data based on the detected action by the other user using the obtained user behavior data; determining, by the computing device, via execution of multiple anomaly prediction models, a plurality of user behavior anomaly predictions, each user behavior anomaly prediction being a different type of anomaly prediction, each of the user behavior anomaly predictions indicating a probability that user behavior represented by the user behavior feature data is anomalous user behavior; aggregating, by the computing device, the different type of the plurality of user behavior anomaly predictions from the multiple anomaly prediction models, the aggregation comprising applying a weight value to a portion of the different type of the plurality of user behavior anomaly predictions; determining, by the computing device, an aggregate user anomaly behavior prediction based on the aggregating of the different type of the plurality of user behavior anomaly predictions; making, by the computing

device, a user behavior anomaly determination using the aggregate user behavior anomaly prediction, the user behavior anomaly determination indicating whether or not the user behavior represented by the user behavior feature data is anomalous user behavior; and providing, by the computing device, functionality for the application to control the action based on the user behavior anomaly determination.

2. The method of claim 1, making a user behavior anomaly determination further comprising: determining, by the computing device, whether or not the user behavior anomaly prediction satisfies a threshold probability.
3. The method of claim 1, wherein the multiple anomaly prediction models comprise an Isolation Forest model, a variational auto-encoder model and a one-class support vector machine.
4. The method of claim 1, wherein the obtained user behavior data is selected from one or more of the following: keyboard usage data, Internet-Protocol (IP) related data, cursor activity data, bot usage data, script usage data, phone sensor data.
5. The method of claim 1, wherein the obtained user behavior data comprises user behavior data in connection with at least one previous session of the user with the application.
6. The method of claim 1, wherein obtaining user behavior data comprises generating additional user behavior data using at least a portion of the user behavior data.
7. The method of claim 6, wherein the additional user behavior data is selected from one or more of the following: funnel navigation pattern, pause length, session length, shortcut usage, script usage, fill form behavior, pointed mouse movement, API calls, local storage data check, aimless mouse movement, device gyroscope parameters, button press length, preference of interaction, information comprehension time, form edit count, and assistance used data.
8. The method of claim 1, generating user behavior feature data further comprising: performing, by the computing device, data preprocessing on a user behavior feature data superset; and selecting, by the computing device, the user behavior data from the user behavior feature data superset in accordance with the data preprocessing.
9. The method of claim 8, wherein the data preprocessing comprises a variance check, correlation analysis and data normalization.
10. The method of claim 8, wherein selecting the user behavior data from the user behavior feature data superset in accordance with the data preprocessing is done using principal component analysis (PCA).
11. The method of claim 1, further comprising: determining, for each of the multiple anomaly prediction models, a probability score related to a user behavior based on the plurality of feature values; combining each probability score respective to a threshold probability value that correlates to a determined number of true and false positives, wherein the aggregation of the determined types of anomaly predictions is further based on the combining of each probability score.
12. A non-transitory computer-readable storage medium tangibly encoded with computer-executable instructions that when executed by a processor associated with a computing device perform a method comprising: obtaining user behavior data in connection with a user of an application; detecting an action by another user related to an interaction with the application; generating user behavior feature data based on the detected action by the other user using the obtained user behavior data; determining, via execution of multiple anomaly prediction models, a plurality of user behavior anomaly predictions, each user behavior anomaly prediction being a different type of anomaly prediction, each of the user behavior anomaly predictions indicating a probability that user behavior represented by the user behavior feature data is anomalous user behavior; aggregating the different type of the plurality of user behavior anomaly predictions from the multiple anomaly prediction models, the aggregation comprising applying a weight value to a portion of the different type of the plurality of user behavior anomaly predictions; determining an aggregate user anomaly behavior prediction based on the aggregating of the different type of the plurality of user behavior anomaly predictions; making a user behavior anomaly determination

using the user behavior anomaly prediction, the aggregate user behavior anomaly determination indicating whether or not the user behavior represented by the user behavior feature data is anomalous user behavior; and providing functionality for the application to control the action based on the user behavior anomaly determination.

13. The non-transitory computer-readable storage medium of claim 12, wherein the multiple anomaly prediction models comprise an Isolation Forest model, a variational auto-encoder model and a one-class support vector machine.

14. The non-transitory computer-readable storage medium of claim 12, generating user behavior feature data further comprising: performing, by the computing device, data preprocessing on a user behavior feature data superset; and selecting, by the computing device, the user behavior data from the user behavior feature data superset in accordance with the data preprocessing.

15. The non-transitory computer-readable storage medium of claim 14, wherein the data preprocessing comprises a variance check, correlation analysis and data normalization.

16. The non-transitory computer-readable storage medium of claim 14, wherein selecting the user behavior data from the user behavior feature data superset in accordance with the data preprocessing is done using principal component analysis (PCA).

17. A computing device comprising: a processor, configured to: obtain user behavior data in connection with a user of an application; detect an action by another user related to an interaction with the application; generate user behavior feature data based on the detected action by the other user using the obtained user behavior data; determine, via execution of multiple anomaly prediction models, a plurality of user behavior anomaly predictions, each user behavior anomaly prediction being a different type of anomaly prediction, each of the user behavior anomaly prediction indicating a probability that user behavior represented by the user behavior feature data is anomalous user behavior; aggregate the different type of the plurality of user behavior anomaly predictions from the multiple anomaly prediction models, the aggregation comprising applying a weight value to a portion of the different type of the plurality of user behavior anomaly predictions; determine an aggregate user anomaly behavior prediction based on the aggregating of the different type of the plurality of user behavior anomaly predictions; make a user behavior anomaly determination using the aggregate user behavior anomaly prediction, the user behavior anomaly determination indicating whether or not the user behavior represented by the user behavior feature data is anomalous user behavior; and provide functionality for the application to control the action based on the user behavior anomaly determination.

18. The computing device of claim 17, wherein the multiple anomaly prediction models comprise an Isolation Forest model, a variational auto-encoder model and a one-class support vector machine.

19. The computing device of claim 17, generating user behavior feature data further comprising: performing, by the computing device, data preprocessing on a user behavior feature data superset; and selecting, by the computing device, the user behavior data from the user behavior feature data superset in accordance with the data preprocessing.

20. The computing device of claim 19, wherein the data preprocessing comprises a variance check, correlation analysis and data normalization and wherein selecting the user behavior data from the user behavior feature data superset in accordance with the data preprocessing is done using principal component analysis (PCA).
